2025 AMC 10B Problem 3 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 3. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 3

Problem:

A Pascal-like triangle has 10 as the top row and 10 followed by 1 as the second row. In each subsequent row the first number is 10, the last number is 1, and, as in the standard Pascal Triangle, each other number in the row is the sum of the two numbers directly above it. The first four rows are shown below.

$$\begin{array}{cccc} 10 & & & \\ 10 & 1 & & \\ 10 & 11 & 1 & \\ 10 & 21 & 12 & 1 \end{array}$$

What is the sum of the digits of the sum of the numbers in the 11th row?

Answer Choices:

A. 11

B. 13

C. 14

D. 16

E. 17

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Solution:

We are given a Pascal-like triangle whose first rows are

$$\begin{array}{llll} 10 & & & \\ 10 & 1 & & \\ 10 & 11 & 1 & \\ 10 & 21 & 12 & 1 \end{array}$$

Each row starts with 10, ends with 1, and each interior entry is the sum of the two numbers directly above it.

Let S_n denote the sum of the numbers in the n-th row (with the top row being $n = 1$). Then

$$S_1 = 10, \qquad S_2 = 10 + 1 = 11.$$

For $n \geq 3$, the n-th row consists of:

$$10,\ (\text{interior entries}),\ 1.$$

Each interior entry is the sum of two entries from the $(n-1)$-st row. When we sum all interior entries, every entry from the $(n-1)$-st row is counted twice. We also notice that the first and the last are also being added twice once to calculate the first and last terms respectively and once for calculating the second and the second last terms respectively.

Therefore, when $n \geq 3$

$$S_n = 2S_{n-1} \implies S_{11} = 2^9 S_2$$

Since, $S_2 = 11$, we get

$$S_{11} = 512 \cdot 11 = 5632.$$

The sum of the digits of 5632 is

$$5 + 6 + 3 + 2 = \boxed{\textbf{(D) } 16}.$$

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